

ERI: Lightweight Learning and Information-State-Driven Planning for Spatiotemporal Environmental Monitoring

1 Research Goals

Surface waters critical for irrigation, drinking, industrial supply, and treatment are increasingly threatened by fuel and oil spills [1], chemical releases [2], and harmful algal blooms [3] (Figure 1). These contamination events create spatiotemporal fields that evolve over hours to days, requiring adaptive, repeated in-situ measurements to protect public health and obtain a synoptic view of aquatic ecosystems. While traditional sampling systems [4, 5] provide valuable but sparse point measurements, a cooperative team of homogeneous autonomous surface vehicles (ASVs) with multimodal sensors can deliver a labor-saving, adaptive, and repeatable data-acquisition capability at the scale of lakes, bayous, and rivers.

However, it remains a significant challenge for existing autonomous robots to collect dense, informative samples across a large, evolving area while operating under onboard compute and energy budgets that are strictly limited. Environmental disturbances such as currents, wind, and waves often impact vehicle speed, causing drift and frequent corrections, while noisy sensor measurements complicate state estimation and mapping. As a result, robots must decide in real time *when*, *where*, and *how* to take samples to best reduce uncertainty while staying within these limits. This is the robotic information gathering (RIG) problem: how a single robot or a team of robots can efficiently maximize information over large, continuous areas within finite mission windows. Informative planning addresses RIG by selecting actions that maximize expected information gain subject to energy, time, and motion constraints. Because robots cannot be teleported, information must be gathered sequentially along physically explored trajectories, and decisions must jointly resolve where to go next and when to revisit nonstationary regions. In practice, field conditions can change within hours (e.g., fronts, inflows, bloom onset), yet many existing methods either rely on heavy offboard computation or assume simple, stationary conditions [6, 7, 8, 9]. These mismatches make it difficult to replan quickly on robotic hardware, avoid redundant sampling, and sustain long-term missions over hours to days.

Furthermore, the central **hypothesis** of this project is that an onboard, information-driven approach—one that runs entirely on the robot, prioritizes the most informative next measurement from a small candidate set, and generalizes to new water bodies and seasons—will enable ASVs to gather the right data at the right places in real time without relying on remote servers. Therefore, two fundamental research questions are posed: **Q1)** How can sampling opportunities be allocated and the next measurement location selected to maximize information gain at onboard speed on resource-limited platforms? **Q2)** How can decision-making strategies be developed to generalize across sites and seasons, remain reliable under changing conditions, and operate within kinematic and operational constraints, all without offboard computation?

The **primary goal of this project** is to develop a low-rank field modeling, limited multi-robot communication, and lightweight, on-device informative decision-making framework so homogeneous robots with multimodal sensors can select and gather in-situ measurements in real time from spatiotemporal fields (e.g., contaminated water sources) and estimate them accurately. At the core of this framework lies the information state [10]—a compact belief (posterior estimate and uncertainty) that is updated online



Figure 1: Examples of contaminated local water sources.

from incoming measurements and used to guide autonomous decision-making. The project will emphasize accuracy under nonstationarity, generalization to new environments, and runtime efficiency on ASVs. To achieve this goal, the project will pursue three **Research Objectives (ROs)**:

- (RO₁) Develop efficient spatiotemporal field modeling and multi-robot coordination methods:** Estimating and monitoring spatiotemporal fields requires compact models that capture both spatial and temporal variations while operating under compute and communication constraints. This project will develop efficient field models and minimal-communication coordination strategies to guide sampling and reduce redundant effort across multiple robots.
- (RO₂) Develop lightweight decision-making strategies for real robot deployment:** Real-time informative sampling in dynamic spatiotemporal fields requires onboard decision-making that acts on a compressed information state under limited computational and energy budgets. This project will synthesize lightweight informative decision-making policies that operate on the information state in real time on robots, enabling them to make safe, resilient, and efficient sampling decisions.
- (RO₃) Validate methods through theoretical analysis, simulations, and field experiments:** The effectiveness of developed methods will be assessed through theoretical analysis, simulations, and physical experiments across diverse environments and conditions. This project will evaluate accuracy, efficiency, and generalization, while demonstrating the feasibility of coordinated multi-robot deployments under resource and communication constraints.

2 Research Project

2.1 Proposed Project in Context of the Existing Work

The RIG problem over environmental fields is typically addressed along two primary lines: (i) information-theoretic planning with probabilistic field models, most commonly Gaussian Processes (GPs) [11, 12, 13, 14, 15, 16, 17], that maximize mutual information (MI) or related submodular objectives; and (ii) trajectory or coverage optimization that shapes continuous paths rather than discrete waypoints to visit regions of high expected information. In the first line, a GP prior captures spatial or spatiotemporal correlations and yields a posterior mean/variance; selecting sample sets or paths to maximize MI is near-optimal under submodularity [18, 19], enabling greedy or recursive-greedy planners and multi-robot approximations. This includes near-optimal observation selection [20] and efficient informative paths [11], and branch-and-bound-driven informative paths [7] that use GP uncertainty directly for path pruning. In the second line, planners optimize continuous trajectories, either via Bayesian optimization [21, 22, 23, 24, 25] or design ergodic trajectories [26] whose time-averaged visitation matches an expected information density, yielding a principled exploration–exploitation tradeoff. Both families have been demonstrated on single- and multi-robot platforms (AUV/ASV/UAV) in simulations and real-world field deployments.

From a modeling perspective, GP inference scales cubically in data size; even with sparsification, online GP map updates and non-myopic look-ahead can dominate cycle time on onboard processors. Spatiotemporal GP variants [27] improve fidelity on evolving fields but add hyperparameters and model-selection overhead, further raising compute and tuning burden. Nonstationary/attentive kernels [28, 29] adapt length-scales to capture fronts and regime shifts, yet require learning local/attention weights and careful regularization, increasing computation and overfitting risk under sparse data. Sparse or online GPs for persistent marine monitoring [30, 31] mitigate cubic scaling but still require inducing-point management and periodic re-optimization, so latency grows as the map expands. Spatiotemporal sequential Bayesian optimization [32] boosts path quality with look-ahead, but non-myopic rollout is combinatorial in horizon and branching factor.

Recent learning-based informative planning methods combine neural networks and reinforcement learning to tackle RIG, providing higher computational efficiency and achieving similar or better planning performance. One representative method [33] combines Monte Carlo tree search (MCTS) with a convolutional neural network (CNN) to learn information-rich actions in adaptive data gathering missions. Others employ recurrent neural networks trained with Q-learning to plan informative paths [34], an attention-based neural network to achieve context-aware path planning [35], and dynamic graph-based policies adapting to evolving maps [36]. A particularly notable recent work [37] presents map-agnostic policies that generalize across diverse map representations by integrating a novel reward function into a representation-agnostic state space.

Across these lines, a recent survey [38] and empirical studies still flag persistent **research gaps**: limited generalization and sim-to-real robustness (performance drops on unseen sites/seasons without substantial domain randomization or adaptation), large offline training budgets with reward-design sensitivity, partial observability and distribution shift that demand belief-state or memory features, and robot deployment shortfalls (desktop-class compute or multi-second replanning). Even where runtime improves, tree-search guidance and non-myopic rollouts remain compute-intensive, and field validations across diverse environments and conditions remain sparse.

2.2 Novel Solution

This Engineering Research Initiation (ERI) project aims to develop a novel learning-based informative planning framework to estimate and track nonstationary spatiotemporal fields with a team of ASVs operating under tight energy and compute budgets. The proposed framework will combine computationally efficient field modeling with learned, on-device informative policies that generalize across environments and transfer from simulation to field. The **fundamental contributions** are as follows:

1. **Low-rank spatiotemporal field modeling** with a linear combination of spatial basis functions, providing a scalable alternative to GP inference.
2. **Information-state updates** that compress observation history into a sufficient statistic, enabling uncertainty- and cost-aware decision-making more efficiently than using the full, high-dimensional state space.
3. **Deployable informative policies** that are trained for generalization and then quantized for real-time execution on resource-constrained ASVs.

2.3 Scientific Challenges Ahead in Executing This Novel Solution and How the PI Will Address Them

Executing this information-state-driven solution poses four intertwined challenges. (1) *Basis selection and dynamics*: Choosing spatial bases that are both expressive and onboard-efficient is nontrivial; purely theoretical bases can mismatch site physics, while fully learned bases risk overfitting and instability. Bases will be selected using domain knowledge and data-driven procedures. A linear system dynamics formulation will enable constant time Kalman filter updates relative to full GP inference [39, 40]. (2) *Generalization and robustness*: Learning-based policies [34, 35, 36] are often sensitive to training distributions with limited transfer to unseen maps/seasons; meta-learning [41] and context-inference RL [42] improve adaptation but are compute-intensive for robot deployments. Training will employ curriculum/domain randomization [43] over procedurally generated maps, with few-shot last-layer adaptation performed on-device [44]. (3) *Runtime guarantees on real robots*: Meeting real-time planning on physical robots requires strict latency control. This will be achieved through small, informative candidate sets, low-complexity I-state updates, and constant-time quantized policies [45, 46], preserving responsiveness on CPU-only platforms. (4)

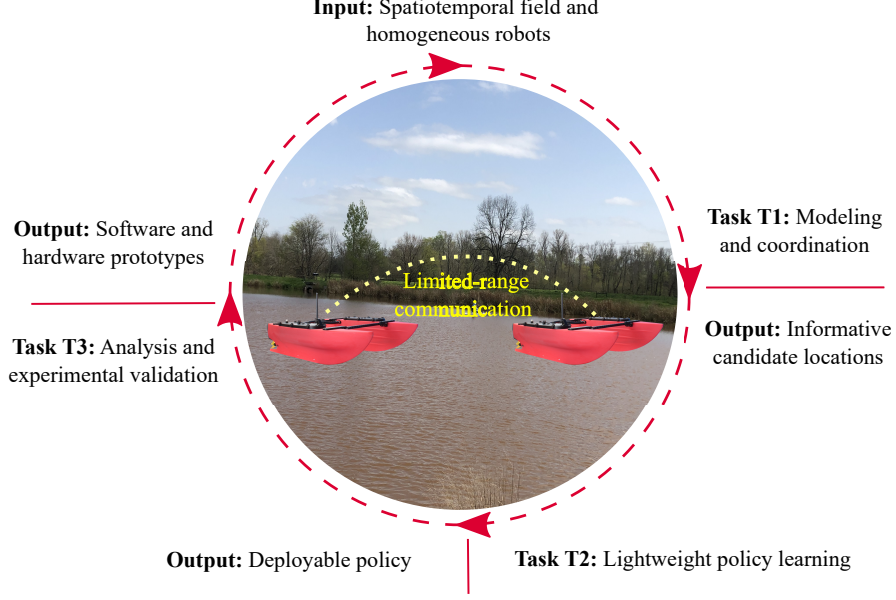


Figure 2: Proposed Research Plan.

Multi-ASV coordination: Field communications are bursty and unreliable, and centralized planning is brittle and bandwidth-heavy. This project will implement asynchronous, limited-range communication protocols based on the PI’s prior work [47]. Together, these measures address known gaps in informative planning for real robots.

2.4 Research Plan

The overall research plan is organized into three interconnected tasks (Figure 2), forming a cyclic pipeline from inputs to software/hardware prototypes. Task T1 develops low-rank spatiotemporal field models and decentralized coordination with limited-range communication to generate a small set of informative candidate locations for sampling. Task T2 learns a lightweight, deployable policy for physical robots that consumes the information state and selects among these candidate locations in real time. Task T3 closes the loop through theory, large-scale simulations, and field experiments, delivering validated software and hardware prototypes.

Task T1: Efficient Spatiotemporal Fields Modeling and Decentralized Coordination - Addresses RO₁

Proposed Research The goal of Task T1 is to construct a scalable, low-complexity model that enables real-time estimation of spatiotemporal fields and supports information-theoretic decision-making by autonomous robots operating under sensing noise, dynamics, and resource constraints. Because environmental processes evolve with uncertainty and robots cannot directly access the full field state, decisions must rely on a compressed belief—an **information state (I-state)** [48, 10]—that captures past observations and their uncertainty. The project will therefore leverage the I-state rather than the full state space to support decision-making.

Low-Rank Field Model and Evolution The scalar dynamic environmental field (see Figure 3) $f(x, t)$ (e.g., temperature, dissolved oxygen, chlorophyll, pollution concentration) will be modeled as a low-rank spatiotemporal process approximated by a linear combination of r spatial basis functions [49]:

$$f(x, t) \approx h(x)^\top a_t, \quad (1)$$

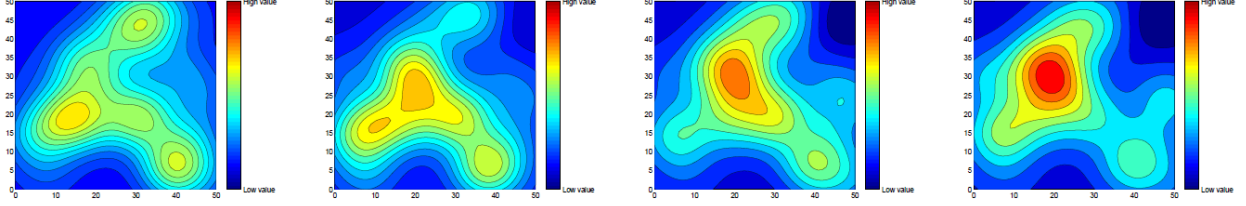


Figure 3: Dynamic spatial field [49] changing from time t_0 to t_3 , where $t_0 < t_1 < t_2 < t_3$.

where $h(x)^\top \in \mathbb{R}^r$ is a vector of spatial basis functions, and $a_t \in \mathbb{R}^r$ is a vector of time-varying weights. The model will provide a compact, expressive representation of spatiotemporal phenomena, with rank $r \ll$ spatial field size. The latent weights will evolve under a linear-Gaussian model with sensor noise in observations.

I-State Updates The I-state will be defined as $\mathcal{I}_t = (a_t, P_t)$ posterior mean of the field weights and their covariance, summarizing all prior observations. I-state updates will be performed using Kalman filtering or recursive least squares (RLS). These updates will yield constant- or quadratic-time complexity per observation—significantly more efficient than the cubic cost of Gaussian process (GP) regression—while maintaining numerical stability on embedded processors [39].

Mutual Information Score For any candidate sensing location u , the posterior variance will be computed as:

$$\sigma^2(u) = h(u)^\top P_t h(u). \quad (2)$$

This variance will be converted into an approximate mutual information (MI) score, which will estimate the expected information gain of observing at location u :

$$\Delta I(u) = \frac{1}{2} \log \left(1 + \frac{\sigma^2(u)}{R_u} \right). \quad (3)$$

This surrogate will preserve the submodularity structure of GP entropy reduction, supporting efficient greedy and recursive-greedy selection [20, 11].

Candidate Generation At each decision time step, a small set U_t of informative, safe candidate locations will be generated using quadtree refinement, frontier-based exploration, or uncertainty sampling heuristics over the current estimated spatiotemporal field, with typical size $K \leq 24$ –32. Each candidate location u will be ranked using a cost-aware utility function $J(u)$ that will balance information gain against costs such as travel distance, heading change, arrival time, and energy.

Decentralized Multi-Robot Coordination When multiple robots operate in the same field, decentralized coordination will be achieved through asynchronous, limited-range communication. Each robot will share compact packets (16–24 bytes) containing its location, estimated arrival time, and marginal mutual information (MI) gain $\Delta I(u)$. Task allocation will be handled using lightweight distributed strategies such as the Consensus-Based Bundle Algorithm (CBBA) or auction-style protocols [50] applied to a shared candidate set U_t . The approximate submodularity of information gain will enable near-optimal task allocation, even under asynchronous communication. This decentralized design builds on the PI’s prior work in asynchronous communication for coordination [47], ensuring robustness to packet loss, delays, and bursty transmission conditions.

Task T2: Learning Lightweight Informative Decision Policies - Addresses RO₂

Proposed Research The goal of Task T2 is to design and train a compact, deployable decision policy that selects among the candidate locations generated in Task T1. This policy will enable autonomous robots to make real-time informative sampling decisions under uncertainty and resource constraints. Unlike traditional policies that map directly from observed states, the proposed policy will operate on the I-state derived in Task T1, which will encode uncertainty, mission context, and safety constraints into compact features. This representation will support safe, efficient, and generalizable decision-making on robotic platforms. The same policy will be executed in a decentralized manner by multiple robots to estimate the underlying field via a linear combination of spatial basis functions, ensuring consistent behavior and coordinated coverage without centralized control.

Policy over I-states At each decision time step, a candidate set U_t of potential sensing locations will be provided by Task T1. For each candidate location $u \in U_t$, an I-state feature vector $\phi_t(u)$ will be constructed as:

$$\phi_t(u) = [\sigma^2(u), \Delta I(u), d(u), \theta(u), \tau(u), e(u), \text{age}(u), \text{trend}(u), \text{safety-bits})]. \quad (4)$$

These features will encode uncertainty (variance $\sigma^2(u)$), mutual information $\Delta I(u)$), mission costs (distance $d(u)$, heading change $\theta(u)$, arrival time $\tau(u)$, energy $e(u)$), temporal context (data age $\text{age}(u)$, recent trend $\text{trend}(u)$), and safety constraints as safety-bits (e.g., depth limit, collision risk).

The policy $\pi(u, \phi_t(u))$ will select the most informative location $u \in U_t$ at each time step from a set of K candidate locations. This will enable real-time, uncertainty-aware decision-making directly from the I-state without requiring access to the full high-dimensional state space.

Imitation Learning and Domain Randomization The learning process will begin with imitation learning from a cost-aware oracle. The policy will first be trained to imitate an oracle that will select actions maximizing a utility function $J(u)$, defined in Task T1:

$$u^* = \arg \max_{u \in U_t} J(u). \quad (5)$$

This imitation phase will establish a strong baseline by aligning the policy with expert decisions.

To enhance generalization and robustness, the policy will be trained in procedurally randomized environments. These environments will include variations in terrain, flow fields, noise levels, and robot initializations. Such domain randomization will ensure that the policy will not overfit to specific maps but instead learn to rely on generalizable I-state features. This approach has proven effective in achieving sim-to-real transfer and enhancing robustness across diverse environments [43, 51, 52], and it will play a key role in ensuring that simulation-trained policies remain effective in real-world deployments. To further mitigate covariate shift, Dataset Aggregation (DAgger) [53] will be employed. During training, the policy will be rolled out in unfamiliar states, and the oracle will be queried to provide corrective feedback. These corrections will be aggregated into the training set and used to iteratively retrain the policy. This feedback loop improves the policy’s robustness, ensuring reliable performance even under out-of-distribution conditions.

Reinforcement Learning Fine-Tuning Following imitation learning, the policy will be fine-tuned with Proximal Policy Optimization (PPO) [54] using a shaped reward function r_t :

$$r_t = \lambda_1 \Delta I(u_t) - \lambda_2 c(u_t) - \lambda_3 \cdot \mathbf{1}_{\text{redundant}} - \lambda_4 \cdot \mathbf{1}_{\text{unsafe}} - \lambda_5 \cdot \mathbf{1}_{\text{noRTD}}, \quad (6)$$

where $c(u_t)$ represents a weighted sum of mission cost components (e.g., distance, time, energy), and the indicator functions apply penalties for redundant actions, unsafe behavior, or failure to return to dock (RTD). This fine-tuning phase enhances the policy’s ability to generalize beyond oracle trajectories, improving robustness under dynamic, partially observed field conditions.



Figure 4: Robots and field site to be used in experimental validation: a) BlueBoat equipped with a YSI multiparameter sonde for in-situ water quality data collection and b) Mission trajectory of the deployed BlueBoat at Pine Island Bayou, Lumberton, TX.

Quantization and Execution To enable real-time execution on low-power, CPU-only robotic platforms, the trained policy network will be compressed and quantized. The policy will be implemented as a compact multilayer perceptron (MLP) or attention-augmented network with fewer than 100,000 parameters, then quantized to 8-bit integers (INT8). This configuration will support inference within 10 milliseconds at under 1 watt of power, staying well within the resource budgets of embedded robotic systems.

In addition to compression, the system will include online adaptation. During execution, residual-based monitoring will detect out-of-distribution conditions. When triggered, a lightweight few-shot update [41] will adjust the final layer using cached intermediate features, completing within approximately 2 seconds to restore performance under shift. To ensure safety, pre-decision filters (e.g., for depth, collision risk, and battery levels) and post-decision checks will be applied to block unsafe actions, thereby ensuring robust and reliable behavior during extended field operations.

Task T3: Theoretical and Experimental Validation - Addresses RO_1 , RO_2 , and RO_3

The goal for Task T3 is to rigorously analyze the theoretical properties of the proposed framework, validate performance at scale in simulation, and demonstrate real-world feasibility through field experiments. This Task will integrate the low-rank field models and decentralized coordination mechanisms from Task T1 with the lightweight, I-state-driven decision policies from Task T2, producing validated software and hardware prototypes. This effort builds upon the experience of PI Alam in testing multi-robot systems [55, 56, 57, 58, 59, 60, 25, 47, 61].

Theoretical Analysis Theoretical justifications will establish conditions under which the proposed framework provides near-optimal performance. In particular, the mutual information (ΔI) from low-rank Kalman filter updates will be shown to approximate Gaussian Process entropy reduction while retaining approximate submodularity, enabling greedy and recursive-greedy planning with bounded optimality guarantees. Coordination performance under decentralized communication will be analyzed using submodular utility theory, and asymptotic regret analysis will quantify long-term efficiency in field estimation.

Preliminary Data Collection To ground the work in real-world conditions, pilot deployments will be conducted using the BlueBoat [62](Figure 4(a)), a compact ASV equipped with a Raspberry Pi-class companion computer, open-source BlueOS, Wi-Fi, flight controller, GPS/IMU, and ArduPilot firmware for autonomous navigation. These deployments will take place at a designated site in Pine Island Bayou, a tributary of the Neches River in Lumberton, TX, with a specified spatial resolution (Figure 4(b)). A YSI EXO multiparameter sonde [63], capable of 1–10 Hz sam-

pling, will be mounted on the BlueBoat to collect in-situ water quality data. The sonde includes a comprehensive suite of sensors measuring depth, temperature, pH, conductivity, salinity, dissolved oxygen, turbidity, chlorophyll, and total algae (phycocyanin), enabling detailed profiling of aquatic environmental conditions. These initial field trials will (i) generate realistic spatiotemporal datasets for simulation design, (ii) characterize sensing noise and variability, and (iii) provide water quality profiles for subsequent field experiments.

Simulations Field-informed simulations will generate ground-truth spatiotemporal fields with realistic sensor noise models, augmented with domain randomization to stress-test algorithms under diverse environments and conditions. A prototype software stack will integrate the BlueBoat simulator [64], ROS2 [65], Gazebo [66], and a Gym-compatible interface [67], enabling large-scale testing of the full framework. Baselines for comparison will include sparse-GP greedy [31], ergodic exploration [26], trajectory Bayesian optimization [21], and recent learning-based informative path planning methods [33, 37]. The evaluation plan will assess the performance across multiple dimensions, including accuracy (RMSE/CRPS), efficiency (time-to-hotspot, duplication rate, computation latency, information-per-joule), cumulative information gain, and cumulative regret. Scalability will also be evaluated by measuring runtime, memory usage, and replanning rate as map size, sample count, and number of robots grow. Ablation studies will isolate the effects of basis rank r , candidate set size K , and policy quantization (e.g., FP32 vs. INT8) on overall performance.

Physical Experiments Finally, multi-day deployments will be conducted across two seasons using two BlueBoats with onboard, quantized policies executing safe, informative decisions in real time. Multi-ASV coordination will rely on asynchronous, limited-range communication, as developed in Task T1. Metrics will mirror those in simulation—mapping accuracy, efficiency, cycle latency, and duplication reduction—while additionally validating safety and adaptation latency under out-of-distribution triggers. This final stage provides end-to-end validation of the integrated framework in realistic aquatic environments.

2.5 Mechanisms and Plans for Assessing Success of the Proposed Activities

With support from this two-year ERI project at Lamar University (LU), PI Alam will mentor one doctoral student in Electrical and Computer Engineering (ECE) and one master’s student in Computer Science (CS) throughout the duration of the award, with an undergraduate CS student added in Year 2. Additional undergraduates will be recruited through the Office of Undergraduate Research and the Texas Academy at LU. The doctoral student will serve as the lead for each research task, supported by the master’s student and mentored directly by the PI. PI Alam will oversee project management and execution, provide domain expertise and existing tools, and ensure consistent, hands-on mentoring at all levels.

2.5.1 Project Timeline

The major research milestones and their anticipated completion periods are summarized in Table 1.

Task	Milestone	Year 1				Year 2			
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
T1	Spatiotemporal Fields Modeling								
T2	Learning Informative Planning Policies								
T3	Theoretical and Experimental Validation								

Table 1: Project timeline of the proposed research plan

2.5.2 Evaluation Plan

The project’s progress will be reviewed against the planned milestones quarterly. The success of the proposed research will be evaluated based on: (1) the timely completion of milestones outlined in Table 1; (2) the number of undergraduate and graduate students participating in this project, their dissertation, thesis, and senior or course projects; (3) the number and quality of peer-reviewed publications and presentations resulting from the proposed activities; and (4) the number of students and teachers engaged through the project’s education and outreach activities.

3 Broader Impacts

This project will significantly advance the capabilities of multi-robot systems in assessing physical, chemical, and biological properties of spatiotemporal fields, thereby enhancing insight into their dynamic behavior. In parallel, targeted educational and outreach activities will reinforce the project’s broader impacts. At the core of these efforts is a dedicated project website, tentatively titled `LightweightRoboticsSolution.com`, which will host collected data, software tools, and tutorials developed throughout the project. The PI proposes five **Broader Impacts Objectives (BIOs)** to be carried out in collaboration with a team of graduate and undergraduate students.

(BIO₁) Developing the Spatiotemporal Fields Estimation Library (SFEL): The PI and his team will release SFEL, an open-source, ROS 2–friendly toolkit for spatiotemporal modeling, informative policy synthesis, and benchmarking. The library will include well-documented APIs, unit tests, tutorials, reproducible baselines, datasets, and continuous integration builds for multiple platforms. To broaden impact and adoption, SFEL will be disseminated through a GitHub organization site, example-ready notebooks, classroom modules for quick course integration, and community events. Evaluation will include GitHub analytics (clones, visitors, referrals, content), engagement signals (stars, forks, issues, time-to-first-response), adoption metrics (pip/Conda downloads, Docker pulls, course or lab integrations), and citations/benchmark submissions.

(BIO₂) Integrating Deep Reinforcement Learning into the CS Curriculum PI Alam will develop a new Deep Reinforcement Learning course, to be offered in both undergraduate and graduate CS programs at LU beginning in Fall 2026. The course will incorporate tutorials and tools from the project website and aim to motivate students to pursue robotics-related research projects. In alignment with the project’s goal, class projects will focus on developing fast, generalizable RL models suitable for deployment on robotic platforms across various applications. Evaluation will include anonymous written student feedback and performance metrics to continuously improve course structure and student engagement.

(BIO₃) Improving Retention of CS Students Amid AI Disruption: Retention of CS students is a central focus of the PI’s mentoring plan. This project will directly address anxiety surrounding AI-driven job displacement by providing structured mentoring for graduate and undergraduate students, hands-on research opportunities, and supportive cohort activities that retain students and motivate them toward advanced study and resilient STEM careers. The project will reframe the prevailing narrative from “AI replacing jobs” to “CS shaping AI,” through a combination of applied projects at the AI–physical interface, career panels with practitioners and alumni, and group discussions of breakthrough AI papers. Evaluation will include tracking changes in student motivation and retention via pre/post surveys, research participation rates, and internship/placement milestones.

(BIO₄) Disseminating Results via a Workshop at a Leading Robotics Conference: Conferences and workshops are ideal for sharing new research findings and increasing the visibility of ongoing research efforts. The PI will present and publish results from this project in top-tier robotics conferences and journals. In addition, the PI will organize a workshop on lightweight reinforcement learning and information-state–driven planning, including related topics aligned with this project’s goal. The proposed workshop will be hosted at a premier robotics venue, such as ICRA or IROS,

and will highlight the project’s technical outcomes, tools, and contributions. The PI will invite attendees from a broad range of respected institutions, including Dr. Leonardo Bobadilla (Florida International University (FIU)), Dr. Dylan Shell and Dr. Jason O’Kane (Texas A&M University), Dr. Aaron T. Becker (University of Houston), Dr. Alberto Quattrini Li (Dartmouth College), and Dr. Ryan Smith (Oxa). This workshop will advance community engagement, highlight the PI’s emerging research direction, and broaden focus on efficient reinforcement learning policies for resource-constrained robotic platforms. Evaluation will include attendee feedback via surveys to assess interests, applicability, and opportunities for collaboration.

(BIO₅) Extending Community Outreach: The PI has a strong track record of outreach through workshops, training, and robotics demonstrations. He has previously taught programming to high school teachers in the Research Experiences for Teachers (RET) program at FIU and organized STEM camps under the Governor’s Summer Merit Program of the Texas Academy at LU, where high school students received hands-on robotics training. Building on this foundation, the PI will invite 10 regional high- and middle-school teachers to LU for workshops on building, programming, and operating robots, enabling them to establish robotics teams at their schools. The PI will also continue summer STEM camps through the Texas Academy, integrating methods developed in this project to engage approximately 40 K–12 students annually in hands-on experiments and demonstrations. Additionally, two robotics demonstration days will be held at local high schools to showcase project outcomes. Evaluation will rely on participant questionnaires and feedback forms to assess impact and inform continuous improvement of outreach activities.

4 ERI Criteria

The PI has research expertise in key topics directly related to this proposal, including informative and feedback planning [68, 69, 59, 25, 47, 70] and planning under uncertainty [71, 72, 73]. The PI has also co-authored publications [55, 56, 74, 75, 76, 77, 78, 79] with scientists and researchers from multiple U.S. universities and institutions that provide well-suited infrastructure for collaboration and validation. Now in the second year of a tenure-track appointment at LU (an R2 research institution), the PI is well positioned to leverage this ERI project to establish an independent and sustained research trajectory. The proposed research lays the foundation for long-term contributions in autonomous environmental monitoring, advancing methods in low-rank field modeling, information-state-driven planning, and resource-efficient multi-robot coordination.

Through this project, the PI will mentor undergraduate and graduate students in CS at LU, engaging them in the development of models, filters, task allocation strategies, coordination methods, and decision-making policies for autonomous surface vehicles (ASVs). This hands-on training will prepare students for both the workforce and advanced research opportunities, enabling them to make sustained scholarly contributions to next-generation engineering research. Addressing the RIG problem under resource and communication constraints—particularly in challenging aquatic environments—will not only advance the state of the art but also strengthen the PI’s capacity for continued scholarly impact.

This applied robotics project provides a solid platform for the PI to expand collaborations with researchers across the nation and to compete for additional federal funding. The outcomes of this work—including theory, algorithms, open-source tools, and field data—will serve as preliminary results for future NSF and DoD proposals, supporting the PI’s long-term goal of developing a comprehensive multi-robot instrumentation system spanning ASVs, UAVs, UGVs, and AUVs for monitoring and decision-making in dynamic environments. By positioning the PI to grow collaborations, mentor students, and generate competitive proposals, this ERI project will directly support his path toward sustained research contributions and leadership in robotics research and education.